English LM fixed. We then fine-tune both English and target model to obtain the bilingual LM. We apply our approach to autoencoding language models with masked language model objective and show the advantage of the proposed approach in zero-shot transfer. Our main contributions in this work are:

- We propose a *fast* adaptation method for obtaining a bilingual $\text{BERT}_{\text{BASE}}$ of English and a target language within a day using one Tesla V100 16GB GPU (§3).
- We evaluate our bilingual LMs for six languages on two zero-shot cross-lingual transfer tasks, namely natural language inference (XNL) (Conneau et al., 2018) and universal dependency parsing. We show that our models offer competitive performance or even better that mBERT (§4).
- We illustrate that our bilingual LMs can serve as an excellent feature extractor in supervised dependency parsing task (§5).

Concurrent to our work, Artetxe et al. (2019) transfer pretrained English model to other languages by fine-tuning only *randomly initialized* target word embeddings while keeping the Transformer encoder fixed. Their approach is simpler than ours but requires more compute (64 TPUv3 chips) to achieve good results.

2 Bilingual Pretrained LMs

We first provide some background of pretrained language models. Let $\boldsymbol{E}_e$ be English word-embeddings and $\text{enc}(\boldsymbol{\theta})$ be the Transformer (Vaswani et al., 2017) encoder with parameters $\boldsymbol{\theta}$. Let $\boldsymbol{e}_{w_i}$ denote the embedding of word w_i (*i.e.*, $\boldsymbol{e}_{w_i} = \boldsymbol{E}_e[w_1]$). We omit positional embeddings and bias for clarity. A pretrained LM typically performs the following computations:

(i) transform a sequence of input tokens to contextualized representations $[\boldsymbol{c}_{w_1}, \ldots, \boldsymbol{c}_{w_n}] = \text{enc}(\boldsymbol{e}_{w_1}, \ldots, \boldsymbol{e}_{w_n}; \boldsymbol{\theta})$

(ii) predict an output word y_i at i^{th} position $p(y_i \mid \boldsymbol{c}_{w_i}) \propto \exp(\boldsymbol{c}_{w_i}^\top \boldsymbol{e}_{y_i})$

Autoencoding LM (Devlin et al., 2019) corrupts some input tokens w_i by replacing them with a special token [MASK]. It then predicts the original tokens $y_i = w_i$ from the corrupted tokens. Autoregressive LM (Radford et al., 2019) predicts the next token ($y_i = w_{i+1}$) given all the previous tokens. The recently proposed XLNet model (Yang et al., 2019) is an autoregressive LM that factorizes output with all possible permutations, which shows empirical performance improvement over GPT-2 due to the ability to capture bidirectional context. Here we assume that the encoder performs necessary masking with respect to each training objective.

Given an English pretrained LM, we wish to learn a bilingual LM for English and a given target language ℓ under a limited computational resource budget. To quickly build a bilingual LM, we directly adapt the English pre-traind model to the target model. Our approach consists of two steps. First, we initialize target language word-embeddings $\boldsymbol{E}_\ell$ in the English embedding space such that embeddings of a target word and its English equivalents are close together (§2.1). Next, we construct a bilingual LM of $\boldsymbol{E}_e$, $\boldsymbol{E}_\ell$, and $\text{enc}(\boldsymbol{\theta})$ and fine-tune all the parameters (§2.2). Figure 1 illustrates the two steps in our approach.

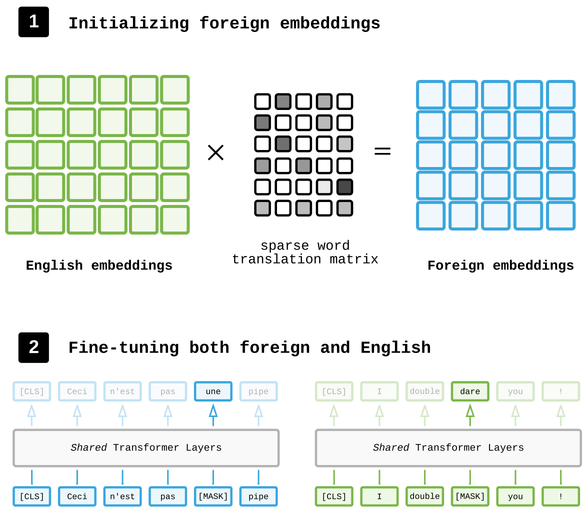

Figure 1: Illustration of our two-step approach. In the first step, foreign embeddings are initialized in English space (§2.1). In the second step, we joinly fine-tune both English and foreign models (§2.2).

2.1 Initializing Target Embeddings

Our approach to learn the initial foreign word embeddings $\boldsymbol{E}_\ell \in \mathbb{R}^{|V_\ell| \times d}$ is based on the idea of mapping the trained English word embeddings $\boldsymbol{E}_e \in \mathbb{R}^{|V_e| \times d}$ to $\boldsymbol{E}_\ell$ such that if a foreign word and an English word are similar in meaning then their embeddings are similar. We represent each foreign